

The Goodhart Gauge: Formal Detection and Mitigation of Metric Manipulation in the Cercis Score Framework

A Formal Analysis of Goodhart’s Law in the SCX Framework

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Abstract

English. Goodhart’s Law states that “when a measure becomes a target, it ceases to be a good measure.” In the SCX egalitarian framework, the Cercis Score $S(x) = Q(x) + \eta N(x)$ aggregates quality Q and novelty N into a single scalar objective. Should this score become an optimization target for self-interested agents—data providers, expert auditors, or state-level actors—the framework is vulnerable to metric manipulation that decouples S from its underlying semantics. This paper formalizes three distinct attack surfaces: (i) *data inflation*, where agents artificially inflate quality metrics without improving genuine quality; (ii) *expert collusion*, where multiple auditors coordinate to fabricate consensus; and (iii) *state forgery*, where adversaries synthesize entire evaluation states to masquerade as authentic Cercis landscapes. For each threat, we prove detection bounds that establish the minimum sample complexity required to distinguish manipulated scores from legitimate ones with high probability. We then propose a layered defense architecture comprising independent audit rotation, state-coverage penalties, and multi-expert independence testing, each accompanied by formal guarantees.

Chinese Abstract. Goodhart’s Law: in the SCX egalitarian framework, the Cercis Score $S(x) = Q(x) + \eta N(x)$ aggregates quality Q and novelty N into a single scalar. Should this score become an optimization target for self-interested agents, the framework is vulnerable to metric manipulation that decouples S from its underlying semantics. This paper formalizes three attack surfaces: (i) **Data Inflation**, where agents artificially inflate quality metrics; (ii) **Expert Collusion**, where multiple auditors fabricate consensus; and (iii) **State Forgery**, where adversaries synthesize entire evaluation states to masquerade as authentic Cercis landscapes.

1 Introduction

1.1 Motivation

Goodhart’s Law, named after the economist Charles Goodhart who observed that “any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes” [?], has become a central concern in the age of AI evaluation. When a scalar metric—be it test-set accuracy, BLEU score, or human-preference rating—becomes the objective that systems or humans optimize against, the metric’s correlation with the underlying construct it was designed to measure degrades, often catastrophically.

Goodhart’s Law in AI. In the age of AI evaluation, Goodhart’s Law poses a fundamental question: can any scalar metric—including the Cercis Score—resist manipulation when it becomes a target? The SCX Equality Principle and the Cercis Score S are designed to be manipulation-resistant, but this paper provides the formal proof.

The SCX egalitarian framework [?] defines the Cercis Score as:

$$S(x) = Q(x) + \eta \cdot N(x), \quad (1)$$

where $Q(x)$ is the intrinsic quality of artifact x , $N(x)$ is its novelty relative to the existing corpus, and $\eta > 0$ is the novelty-weighting parameter. The score lives on the Situs manifold $(\mathcal{X}, d_P, \mu_P)$, where d_P encodes perceptual distance and μ_P encodes the empirical distribution of artifacts.

The Goodhart Threat. Consider an adversary who can observe (or estimate) the Cercis Score $S(x)$ and wishes to maximize it without incurring the cost of genuine quality improvement. Three canonical attack strategies emerge:

A1 Data Inflation (A1). The adversary generates artifacts $\tilde{x}$ that score highly on S by exploiting blind spots in the Q estimator, without improving true quality. This is the analog of “teaching to the test.”

A2 Expert Collusion (A2). Multiple auditors $\{E_k\}_{k=1}^K$ collude to report inflated or coordinated scores that mimic genuine inter-rater disagreement patterns while systematically shifting the consensus upward.

A3 State Forgery (A3). The adversary synthesizes an entire evaluation state—a tuple $(x, S(x), \nabla S(x), H_S(x))$ —that is indistinguishable from authentic Cercis states to the downstream consumer (model trainer, policy maker, funding agency).

1.2 Contributions

This paper makes the following contributions:

C1 Threat Formalization (Section ??). We provide rigorous mathematical definitions of A1–A3 within the SCX framework, specifying the adversary’s capabilities, knowledge, and objective for each attack surface.

C2 Detection Bounds (Section ??). For each attack, we prove a detection theorem that quantifies the minimum number of samples n required to reject the null hypothesis of “no manipulation” at significance level α with power $1 - \beta$. The bounds expose an inherent detection-manipulation trade-off governed by the Cercis novelty parameter η .

C3 Countermeasure Architecture (Section ??). We propose three defense mechanisms with formal guarantees: (a) independent audit rotation, which decorrelates auditor assignments to break collusion equilibria; (b) state-coverage penalties, which penalize concentration of evaluated artifacts in regions of Situs space; and (c) multi-expert independence testing, which detects collusion via conditional independence violations.

C4 Bilingual Exposition (throughout). All theorems, definitions, and interpretations are presented in both English and Chinese to serve the international SCX research community.

2 Threat Model

Threat Model Summary. The table below summarizes the three attack surfaces, specifying adversary capability, objective, and the structural vulnerability exploited:

Attack	Capability	Objective	Vulnerability	Detection
A1: Data Inflation	Proxy Q access	$\max S(\tilde{x})$	Q estimator gap	Score distribution shift
A2: Expert Collusion	Coordinated reporting	$\max \bar{S}_{\text{rep}}$	Auditor dependence	Correlation anomaly
A3: State Forgery	Full SCX knowledge	$\max S(x)$	$\Sigma(x)$ geometry	Curl inconsistency

2.1 A1: Data Inflation

Definition 2.1 (Data Inflation Attack). An adversary $\mathcal{A}_1$ with access to a surrogate quality estimator $\hat{Q}(\cdot)$ produces artifacts $\{\tilde{x}_i\}_{i=1}^m$ satisfying:

$$\hat{S}(\tilde{x}_i) = \hat{Q}(\tilde{x}_i) + \eta \hat{N}(\tilde{x}_i) \geq \tau,$$

for some threshold τ , while

$$\mathbb{E}[Q(\tilde{x}_i)] \leq \mathbb{E}[\hat{Q}(\tilde{x}_i)] - \Delta_Q,$$

where $\Delta_Q > 0$ is the **quality gap** between estimated and true quality.

Core mechanism. The gap between proxy Q and true Q creates the attack surface. The adversary exploits the estimator’s blind spots—what the proxy model $\hat{Q}$ scores highly, but the true Q does not.

The quality gap Δ_Q is the adversary’s “Goodhart margin”—the extent to which the proxy metric $\hat{Q}$ overestimates true quality Q . This gap arises from:

- (i) **Estimator incompleteness:** $\hat{Q}$ captures only a subset of the quality dimensions that Q measures.
- (ii) **Distribution shift:** $\hat{Q}$ was calibrated on a training distribution $\mathcal{D}_{\text{train}}$ that differs from the adversary’s generation distribution $\mathcal{D}_{\text{adv}}$.
- (iii) **Adversarial overfitting:** the adversary actively searches for inputs where $|\hat{Q}(x) - Q(x)|$ is large.

2.2 A2: Expert Collusion

Definition 2.2 (Expert Collusion Attack). Let $\mathcal{E} = \{E_1, \dots, E_K\}$ be K expert auditors assigned to evaluate artifact x . In a collusion attack, a subset $\mathcal{E}_C \subseteq \mathcal{E}$ of size $K_C \leq K$ coordinates to report:

$$S_k^{\text{rep}}(x) = S(x) + \delta + \tilde{\varepsilon}_k,$$

where $\delta > 0$ is a **collusive bias** (common across colluders) and $\tilde{\varepsilon}_k \sim \mathcal{N}(0, \tilde{\sigma}_k^2)$ are fabricated noise terms calibrated to match the variance of honest auditors.

Colluder strategy. The colluders face two objectives: (1) inflate the aggregate Score ($\delta > 0$); (2) mimic honest disagreement ($\tilde{\varepsilon}_k$ calibrated to match σ_k). The colluders’ dilemma is that δ cannot be made arbitrarily large without being detected by outlier tests, but must be large enough to meaningfully shift the aggregate score $\bar{S}$.

2.3 A3: State Forgery

Definition 2.3 (State Forgery Attack). An adversary $\mathcal{A}_3$ with complete knowledge of the SCX framework produces forged states $\tilde{\Sigma}(x) = (\tilde{x}, \tilde{S}, \tilde{\nabla} S, \tilde{H}_S)$ such that the distribution $\mathcal{D}_{\text{forge}}$ of forged states is ε -close to the distribution $\mathcal{D}_{\text{legit}}$ of legitimate states in total variation:

$$d_{\text{TV}}(\mathcal{D}_{\text{forge}}, \mathcal{D}_{\text{legit}}) \leq \varepsilon.$$

State forgery. A3 is the most sophisticated attack: the adversary forges not just the Score, but the entire Cercis State including gradient and Hessian, making the forgery geometrically consistent with the Situs manifold.

3 Detection Bounds

Detection framework. For each attack, we prove a detection theorem specifying the minimum sample size n^* required to achieve significance level α and power $1 - \beta$. The bounds reveal how detection difficulty scales with manipulation magnitude $(\Delta_Q, \delta, \varepsilon)$ and the Cercis parameter η .

3.1 Detecting Data Inflation

Theorem 3.1 (Data Inflation Detection Bound). *Let $\mathcal{D}_{\text{legit}}$ be the distribution of Cercis Scores for authentic artifacts with mean μ_S and variance σ_S^2 . Let $\mathcal{D}_{\text{infl}}$ be the distribution of inflated artifacts with mean $\mu_S + \Delta_{\text{infl}}$ and variance $\sigma_S^2 + \Delta_\sigma^2$. Assume $\Delta_Q = \Delta_{\text{infl}} - \eta \cdot \Delta_N$, where $\Delta_N \geq 0$ is the novelty shift. Then:*

- (i) **Score-based detection.** *A two-sample t -test on n observed scores detects inflation at level α with power $1 - \beta$ when:*

$$n \geq n_{A1}^* = \frac{2(\sigma_S^2 + \Delta_\sigma^2/2)}{\Delta_{\text{infl}}^2} \cdot (z_{1-\alpha/2} + z_{1-\beta})^2. \quad (2)$$

- (ii) **Gradient-based detection.** *If the adversary’s artifacts exhibit anomalous gradient structure $\|\nabla S(\tilde{x})\| \leq G_{\text{legit}} - \Delta_G$ then detection is possible with:*

$$n \geq n_{A1,\text{grad}}^* = \frac{2\sigma_G^2}{\Delta_G^2} \cdot (z_{1-\alpha/2} + z_{1-\beta})^2, \quad (3)$$

where $\sigma_G^2 = \mathbb{V}[\|\nabla S(x)\|]$ under the legitimate distribution.

Proof. Part (i). Under H_0 (no inflation), scores are i.i.d. with mean μ_S . Under H_1 (inflation), scores have mean $\mu_S + \Delta_{\text{infl}}$. By the Central Limit Theorem, the test statistic is asymptotically normal. The standard sample-size formula for a two-sided z -test with pooled variance $\sigma_{\text{pool}}^2 = \sigma_S^2 + \Delta_\sigma^2/2$ yields (??).

Part (ii). Same logic applied to the gradient norm statistic $G(x) = \|\nabla S(x)\|$. The separation Δ_G captures the fact that inflated artifacts tend to lie in “flat” regions of S (local optima of the proxy) rather than the “rugged” regions characteristic of genuinely novel, high-quality artifacts. $\square$

Remark 3.2 (Interpretation). **The role of η .** The sample size n_{A1}^* depends on:

- Δ_{infl} (inflation magnitude): larger inflation $\rightarrow$ easier detection;
- Δ_σ^2 (inflation increases variance): more variance $\rightarrow$ harder detection;
- η enters via $\Delta_{\text{infl}} = \Delta_Q + \eta\Delta_N$: when novelty is inflated ($\Delta_N > 0$), larger η amplifies the inflation signal, making detection easier. **Self-limiting property: larger η makes the Cercis Score more valuable but also makes its manipulation more detectable.**

3.2 Detecting Expert Collusion

Theorem 3.3 (Expert Collusion Detection Bound). *Let K experts evaluate artifact x , of whom $K_C \leq \lfloor K/2 \rfloor$ are colluding with common bias δ and calibrated variance $\sigma_C^2 = \sigma_H^2$, where σ_H^2 is the honest-expert variance. Define the **inter-expert correlation anomaly**:*

$$\rho_{\text{anom}} = \frac{\delta^2}{\delta^2 + 2\sigma_H^2}.$$

Then:

(i) **Pairwise correlation test.** For any pair of colluding experts (i, j) , a z -test on Fisher-transformed correlations detects collusion when:

$$n \geq n_{A2, \text{corr}}^* = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2}{\left(\frac{1}{2} \ln \frac{1+\rho_{\text{anom}}}{1-\rho_{\text{anom}}}\right)^2} + 3. \quad (4)$$

(ii) **Variance ratio test.** Define the **collusion F -statistic**:

$$F_{\text{coll}} = \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{K_C(K - K_C)\delta^2 + \sigma_H^2}{\sigma_H^2}. \quad (5)$$

Detection at level α requires:

$$n \geq \frac{(\sigma_H^2/\delta^2) \cdot (F_{1, K-2; 1-\beta} - 1)}{K_C(K - K_C)}. \quad (6)$$

3.3 Detecting State Forgery

Theorem 3.4 (State Forgery Detection Bound). Let $\mathcal{D}_{\text{legit}}$ be the joint distribution of Cercis states $\Sigma(x) = (x, S, \nabla S, H_S)$ for legitimate artifacts. Define the **curl inconsistency statistic**:

$$\tau(x) = \|\nabla \times \nabla S(x)\|^2, \quad (7)$$

where $\nabla \times$ is the exterior derivative on the Situs manifold. A one-sided z -test on $\bar{\tau}_n$ detects forgery when:

$$n \geq n_{A3}^* = \frac{\sigma_\tau^2}{\Delta_\tau^2} \cdot (z_{1-\alpha} + z_{1-\beta})^2, \quad (8)$$

where $\sigma_\tau^2 = \mathbb{V}_{\text{legit}}[\tau(x)]$.

Additionally, forged states must satisfy the **Cercis-Lipschitz constraint**:

$$|\tilde{S}_i - \tilde{S}_j| \leq (L_Q + \eta L_N) \cdot d_P(\tilde{x}_i, \tilde{x}_j). \quad (9)$$

Violations of this constraint constitute detectable anomalies. The probability of a random pair violating the constraint under H_0 is bounded by:

$$\mathbb{P}_{H_0}(\text{violation}) \leq 2 \exp \left(-\frac{(L_Q + \eta L_N)^2 \cdot d_{\min}^2}{2\sigma_S^2} \right),$$

where $d_{\min} = \min_{i \neq j} d_P(\tilde{x}_i, \tilde{x}_j)$.

3.4 Unified Detection Frontier

Theorem 3.5 (Goodhart Detection Frontier). For all three attacks, the minimum sample size for detection scales as:

$$n^*(\Delta, \alpha, \beta) = \Theta \left(\frac{1}{\Delta^2} \right), \quad (10)$$

with constants depending on the specific attack structure and Cercis parameter η . Furthermore, the **detection possibility region** is:

$$_{\text{detect}} = \left\{ (\Delta, n) : n \geq \frac{C(\eta, \sigma)}{\Delta^2} \right\},$$

where $C(\eta, \sigma)$ is a constant that:

- (i) Decreases with η for A1 (novelty amplifies inflation signals);
- (ii) Increases with η for A2 (novelty-weighting increases score variance, masking collusion);
- (iii) Is independent of η for A3 (curl inconsistency depends only on geometric structure).

4 Countermeasures

Defense architecture. We propose three layered countermeasures, each with formal guarantees.

4.1 Countermeasure 1: Independent Audit Rotation

The core insight is that collusion requires *repeated interaction*. Random rotation breaks this.

Definition 4.1 (Audit Rotation Protocol). Let $\mathcal{E} = \{E_1, \dots, E_M\}$ be the pool of M certified experts. For each artifact x_t , an **independent audit panel** $_t \subset \mathcal{E}$ of size K is drawn uniformly at random without replacement.

Theorem 4.2 (Rotation Breaks Collusion Equilibria). *Under the rotation protocol, any collusion strategy requiring pre-commitment of ≥ 2 auditors has expected coordination probability:*

$$\mathbb{P}(\text{successful coordination}) \leq \binom{K_C}{2} / \binom{M}{2} \leq \frac{K_C^2}{M^2},$$

which decays quadratically in the pool size M .

4.2 Countermeasure 2: State Coverage Penalty

Data inflation attacks concentrate artifacts in “easy-scoring” regions. A coverage penalty discourages this.

Definition 4.3 (Coverage-Adjusted Cercis Score). Let $p(x)$ be the empirical density of evaluated artifacts at $x \in \mathcal{X}$. Define the **coverage-adjusted Cercis Score**:

$$S_{\text{cov}}(x) = S(x) - \lambda \cdot \log p(x), \quad (11)$$

where $\lambda > 0$ is the coverage penalty parameter.

4.3 Countermeasure 3: Multi-Expert Independence Testing

Online monitoring detects collusion via conditional independence violations.

Definition 4.4 (Online Independence Monitor). For each artifact x , the **independence monitor** computes:

M1 Pairwise Correlation Matrix over a sliding window of n artifacts.

M2 Conditional Independence Test: $\text{CI}_{ij|S} = \text{partial correlation}(S_i, S_j \mid \hat{S})$.

M3 CUSUM Anomaly Detector: $C_t = \max(0, C_{t-1} + \log \Lambda_t - \gamma)$.

5 Composition with SCX Theorems

The Goodhart defenses compose with existing SCX theorems to provide end-to-end guarantees:

Theorem 5.1 (Defense Composition Theorem). *The Goodhart countermeasures $\mathcal{G} = (\text{Rotation}, \text{Coverage}, \text{Independence})$ compose with the SCX theorem suite as follows:*

- (i) **With Theorem 3 (Noise-Difficulty):** $\mathcal{G}$ adds a geometric detection layer orthogonal to semantic indistinguishability.
- (ii) **With Theorem 5 (Active Learning):** Coverage penalty modifies the Cercis-Gain function, improving diversity guarantees.
- (iii) **With Theorem 7 (Cross-Domain):** Situs-Lipschitz constraint provides bounds on cross-domain transfer fidelity.

6 Conclusion

Conclusion. Goodhart’s Law is not a fatal vulnerability of the Cercis framework, but rather its *detectable* degradation mode. Because the Cercis Score is defined with geometric structure (the Situs manifold) rather than as a bare scalar, manipulation leaves detectable traces in the geometry—gradient inconsistencies, collusion-induced conditional correlations, and anomalous concentration of evaluation density.

The measure is information: when the measure is manipulated, the act of manipulation itself generates new information that can be used to detect the manipulation. This is the Cercis framework’s fundamental response to the Goodhart dilemma.

References

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